A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 01

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** A: Student's t-test - a difference between the means of two sets of normally distributed data **1** B: chi-squared test - a comparison of observed frequencies (counts) with expected frequencies **1** C: Spearman's rank correlation - a relationship between two variables measured on the same sampling points **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Student's t** → a test for a difference between the means of two normally distributed sets • **chi-squared** → a test comparing observed frequencies with expected frequencies • **Spearman** → a test for correlation between two sets of ranked data **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 2% probability, or a 1 in 50 chance, that a result at least as extreme as this would arise by chance alone if the null hypothesis were true **1** This is below the 5% significance level normally used, so the null hypothesis is rejected and the difference or correlation is described as statistically significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **significance level** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** The calculated value on its own has no meaning: it must be compared with the value that would be exceeded by chance in 5% of cases, which is read from a table of critical values **1** The critical value depends on the number of degrees of freedom, which comes from the number of categories or pairs of data, because a larger sample makes a given result less likely to be chance **1** If the calculated value exceeds the critical value at p = 0.05 the null hypothesis is rejected – if it does not, the null hypothesis is not rejected and the result is not significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **critical value** → the table value the calculated statistic must reach to be significant • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3

Do not write outside the box

Question	Marking guidance	Marks
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** The standard deviation measures the spread of the individual values about the mean, so a small value means the heights were closely grouped and the mean is a good summary of them **1** Comparing the standard deviations with the difference between the means shows whether the two sets really are distinct or merely overlap **1** It is drawn as error bars above and below each plotted mean – where the error bars of two means do not overlap, a significant difference is likely, and where they overlap widely it is not **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	3
01.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** A correlation shows only that the two variables change together – it suggests a link that further work would have to test **1** A third factor may cause both, for instance shade, which raises soil moisture by reducing evaporation and also directly favours the moss **1** The relationship may run the other way round: a dense moss cover holds water and shades the soil, so the moss could be raising the moisture rather than responding to it **1** The points along one transect are close together and not independent, so a pattern may reflect one gradient across the site rather than a general relationship **1** Only an experiment in which soil moisture is changed while every other variable is held constant can establish cause **1** A result significant at p = 0.05 will still arise by chance about 1 time in 20, so a single significant result is not proof **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together • **transect** → a line along which samples are taken to show change with distance • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407, 420-423; AQA spec 3.7.2.3.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **Critical Group** → the people most at risk / most heavily exposed	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate **1** No need to sample sites that were never likely to be polluted – sampling also checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify **1** So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** The amount of energy contained in a given mass of the resource (or harnessed per kg of equipment for renewables) **1** High energy density means smaller quantities are needed so storage and transport are easier and cheaper (eg 75 t of aviation fuel carries a 400 t aircraft 5600 km) **1** High energy density fuels reach high temperatures needed for smelting / steam turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 500 000 / 28 = 17 857 kg **1** = 17.9 tonnes (accept 18; book quotes 13 t for 0.7% enriched fuel) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** The electricity they generate can power high-energy-requirement technologies **1** Electricity can be converted to high energy density secondary fuels, eg hydrogen by electrolysis, or stored (batteries, pumped storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1 Intermittent:** not available all the time, eg tidal (but the times are predictable so alternatives can be planned) **1 Unpredictable:** intermittent and the timing / amount cannot be forecast accurately, eg wind, solar **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • Any appropriate named example, including ones that are not listed here. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2
03.5	1 mark for the point below. **1** Gaseous natural gas has a very low energy density by volume – LNG at -160 °C has 160 times the energy density so a ship can carry a worthwhile amount **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-178; AQA spec 3.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Water = 65.0 - 55.0 = 10.0 g – fresh soil = 40.0 g **1** 10.0 / 40.0 x 100 = 25.0% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Organic matter = 55.0 - 52.6 = 2.4 g – dry soil = 30.0 g **1** 2.4 / 30.0 x 100 = 8.0% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** So all water is removed and the mass loss on burning is due to organic matter only **1** Heating until mass no longer changes shows all water has evaporated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** Organic matter releases nutrients as it decomposes and provides food for soil biota **1** Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration **1** Surface organic matter / humus adhesion reduce rain splash and wind erosion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **peds** → soil crumbs / aggregates of soil particles	3

Do not write outside the box

Question	Marking guidance	Marks
04.5	1 mark for the point below. **1** Samples taken from different depths / locations (organic matter is concentrated near the surface) – incomplete burning – different heating times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** At times of low demand / surplus supply (night, windy periods) surplus electricity pumps water from the lower to the upper reservoir, storing it as gravitational potential energy **1** At peaks in demand water flows down through the turbines generating electricity **1** Can go from standby to full power in under 15 seconds so it meets sudden peaks (TV pickup) that fossil / nuclear stations cannot follow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Mass = $7 \times 10^6 \times 1000 = 7 \times 10^9$ kg **1** PE = m g h = $7 \times 10^9 \times 9.8 \times 500$ **1** = 3.43×10^{13} J (accept 3.4×10^{13}) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (J) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** The electricity used for pumping is surplus that would otherwise be wasted / lost as heat, or cheap off-peak electricity **1** It avoids building rapid-response low-efficiency peaking plants (open-cycle gas turbines) and allows intermittent renewables to be used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
05.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Two water bodies / sites with a large height difference close together (mountainous) **1** Impermeable, stable rock **1** Close to the grid / demand **1** Adequate water supply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222; AQA spec 3.3.4.1.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Organic (methyl) mercury **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Anaerobic microbes converted the inorganic mercury into organic methyl mercury **1** Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are less soluble at high pH **1** So they remain immobile and are not leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** Trade winds blow westwards across the Pacific near the equator, driving a surface current in the same direction **1** Surface water moving away from South America is replaced by deep, cold water drawn upwards (upwelling) near the coast **1** The upwelling water is rich in nutrients (nitrates, phosphates) so large algal / phytoplankton blooms occur **1** These support a rich food web including the anchovy fishery – the water warms as it travels west **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	4
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** The trade winds weaken or reverse so the surface current slows / reverses and warm water spreads east **1** Upwelling of nutrient-rich cold water off South America stops **1** Phytoplankton productivity collapses so the food web / anchovy population collapses – catch fell from about 7.7 to 1.7 million tonnes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Heavy rain and floods in the normally dry coastal deserts of Peru / South America **1** Drought and reduced rainfall in eastern Australia / Indonesia **1** Droughts in NE Africa, southern Africa, China **1** Fewer hurricanes in the North Atlantic / fewer cyclones in Japan **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
07.4	1 mark for the point below. **1** Events appear to be occurring more frequently and are driven by wind patterns that warming may alter, but the causes of ENSO are not fully understood and natural fluctuations make trends hard to separate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-99; AQA spec 3.2.1.2, 3.2.2.3.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions **1** The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.2	**1 mark each** 1 mark for each of the 2 points below. **1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) **1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	1 mark for the point below. **1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Total = 42 + 26 + 31 + 9 + 22 = 130 kW **1** 31 / 130 x 100 = 23.8% (accept 24%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
09.2	**1 mark each** 1 mark for each of the 3 points below. **1 Cavity wall insulation:** the foam or fibre traps air in small pockets so convection in the cavity is prevented and the low thermal conductivity of still air reduces conduction through the wall **1 Low-emissivity double glazing:** the still air or inert gas between the panes conducts heat poorly, and the low-emissivity coating reflects infrared radiation back into the room instead of letting it radiate out **1 Heat exchanger:** outgoing stale warm air passes next to, but does not mix with, incoming cold fresh air, so most of the heat is transferred to the incoming air and the building can be ventilated without losing that heat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each extra layer saves less heat than the one before, because the layers already in place carry most of the temperature difference, so the saving per unit thickness falls **1** Beyond a certain thickness the cost and the embodied energy of making and fitting more insulation are never repaid by the energy it saves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 3 points below. **1** A large power station runs at higher temperatures and pressures and uses large efficient turbines, so it converts a greater proportion of the fuel energy into electricity **1** The heat rejected by a large station is low-grade and the station is usually too far from any user, so it is lost to a river or cooling tower **1** A CHP unit is in the building that needs the heat, so the heat rejected by its generator is piped straight to the heating and hot water, and 70 - 80% of the fuel energy is used rather than about 35% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **combined heat and power** → using the waste heat from generation for heating	3
09.5	**Any 5 of 6** 1 mark each for any 5 of the 6 points below, in any order, maximum 5. **1** Occupancy sensors and programmable thermostats so that lighting and heating run only where and when people are present **1** LED lighting, which converts a far greater proportion of the electricity into light and much less into waste heat than filament lamps **1** Large south-facing windows and a thermally massive structure to make use of passive solar gain, with shading to prevent overheating in summer **1** Heating water on demand at the point of use, so no energy is wasted keeping a tank hot and none is lost in the pipe run **1** Choosing materials with low embodied energy, and re-using the existing structure rather than demolishing and rebuilding **1 Changes in behaviour:** turning equipment off rather than leaving it on standby, closing doors, and setting the thermostat a degree lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-235; AQA spec 3.3.4.2.2-3.3.4.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Neonicotinoids are neurotoxins similar to nicotine – they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • **Very toxic to bees:** sub-lethal doses reduce navigation ability, foraging and immunity to disease – linked to colony decline • **Water soluble and moderately persistent:** remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil – broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • **Control:** seed coating rather than spraying – spraying when crops are not in flower / at night – still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops) – use of alternative pest control (cultural, biological, integrated control) • **Evaluation:** reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators – trade-off with crop yields **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **persistent** → does not break down / stays in the environment for a long time • **Synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit**	9
	• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Neonicotinoids are neurotoxins related to nicotine that block the neurotransmitter acetylcholine. They are highly toxic to insects but of low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier - the specificity that made them popular. Bees, however, are insects: even sub-lethal doses impair navigation, foraging and immunity, and this has been linked to colony decline. Because the compounds are water-soluble and moderately persistent they remain in the soil, are absorbed by roots and translocated through the plant (systemic action), so they appear in pollen and nectar that bees collect; persistence may also allow build-up in soils and aquifers, although sunlight and soil microbes break them down. Some fungicides act synergistically with them, increasing toxicity to bees.	
	Exposure is greatest when crops are sprayed or when dust from coated seeds is blown by wind, so impacts are reduced by using seed coatings instead of sprays, spraying only when crops are not in flower or at night when bees are not foraging, and applying on still days to limit drift. The EU has banned outdoor use on flowering crops, and integrated control with cultural and biological methods reduces the need for insecticides altogether.	
	These measures reduce bee exposure, but the alternatives farmers turn to, such as pyrethroids, also harm pollinators, and restrictions can reduce yields, so the trade-off between crop protection and pollination services remains.	
	Weaker answer - Level 2 (4 marks)	
	Neonicotinoids are insecticides that kill insects by affecting their nervous system. They are systemic so they get into the whole plant including the pollen, and bees pick them up when they collect pollen and nectar. This harms bees and can cause colonies to die. To reduce the impact, farmers can spray when crops are not flowering, and some neonicotinoids have been banned in the EU. Other methods of pest control could be used instead.	
	What makes the difference	
	• Level 2 links the systemic property to bee exposure correctly but omits the other properties (solubility, persistence, specificity via the blood-brain barrier, synergism) that the question's 'properties' demands.	
	• Level 3 explains the sub-lethal effects (navigation, foraging, immunity) rather than 'harms bees'.	
	• Level 3 includes the trade-off evaluation (alternatives also harm pollinators, yield costs) that the mark scheme rewards.	

Do not write outside the box

Question	Marking guidance	Marks
10.2	**Any 1 of 6** 1 mark each for any 1 of the 6 points below, in any order, maximum 1. **1** Biological control (introduced predators / pathogens) **1** Predator habitats (hedgerows, beetle banks) **1** Crop rotation **1** Companion / barrier crops **1** Sterile male technique / pheromone traps **1** Genetic resistance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Stages of investigation:** identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • **Why Antarctica:** low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • **Effects of UVB:** skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • **Solutions:** Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • **Evaluation of success:** ozone hole area / depth stabilising, CFC levels falling	25
	– limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier – role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	

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Question	Marking guidance	Marks
11.2	**Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Nature of the problem:** interconnected atmosphere-hydrosphere-biosphere systems – changes over different time and spatial scales – natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change – time delay between cause and effect (ocean heat capacity) • **Limitations of historic data:** sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • **Proxy data and their reliability:** dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • **Modern monitoring:** long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty – ecological uncertainty (species survival, distribution, fragmentation) • **Computer models:** how they are tested and refined – importance of accurate representative data – scenario-based predictions and ranges	25
	• Uncertainty over ENSO / Conveyor / jet stream responses • **Evaluation and conclusion:** confidence in the overall trend and human cause despite detailed uncertainty – role of peer review and international bodies (IPCC) *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	

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